

Tools that Help to Secure and Monitor Infrastructure as a Service

Due to its many advantages and conveniences, cloud computing has become the norm rather than the exception for enterprises. In such a scenario, the security and monitoring of cloud services becomes extremely important. This article looks at how to secure Infrastructure as a Service (IaaS).



Infrastructure as a Service

Before getting into the security aspects of IaaS, let's first understand what it is all about and why securing it is more important than protecting other such services like PaaS and SaaS.

Infrastructure as a Service is the base for Platform as a Service (PaaS) and Software as a Service (SaaS), as these services are built on top of IaaS. Visualise IaaS as being the basement of a building, which is why it is very important to build a secure IaaS service to provide secured PaaS and SaaS services. In general, IaaS consists of compute, storage and network infrastructure, which you can quickly scale up and down, based on demand.

As part of IaaS, most vendors provide virtual machines, storage, serverless computing and containers.

IaaS security challenges and solutions

Insider threats: Such threats are the biggest cyber security problem for organisations nowadays. They are harder to detect and prevent in comparison to attacks from the outside. Insiders know where the sensitive company and customer data is, and they have access to it. That means insiders know exactly where to strike if they decide to take action. This is why insider attacks have a much bigger negative impact on business.

Orphan disks and images: These are storage disks that are not attached to any computing servers. Improper handling of these disks can lead to leakage of sensitive data to the outside world. Even the misuse of these disks can go unnoticed for a long time. Unused machine images and object storage may have sensitive data, which should be deleted or moved to archival storage with proper access controls.

Account management gets compromised: Account management includes provisioning new users, managing user credentials, maintaining required access privileges, defining roles and the right set of policies. These processes are critical when it comes to

